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Startup IP Is the Most Underpriced Asset on Your Cap Table

The right always existed. What protected the conduct around it was the cost of finding out — and that cost is collapsing. The advice you are still getting is twenty years old, and it has moved from fact into myth

Kenneth Lawrence Cheney

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The right always existed. What made it safe to ignore was the cost of finding out — and that cost is collapsing. The advice you are still getting is twenty years old: arithmetic when it was written, myth by the time it reached you.

Kenneth L. Cheney · Expound Laboratories™ · August 2026

A follow-on to Who Can Afford Friction?

That article argued artificial intelligence changes *who can afford to exercise a right*.

This one applies it to the rights I hold, and to the ones you are being told not to bother with.

Before you read on

This article has two audiences and it is not saying the same thing to each of them. Here is what each should take away, stated up front so nobody has to infer it.

If you are a founder or run a startup:

1. **You have been pricing the wrong event.** The number you were quoted is the cost of a full trial through appeal in a very large dispute. Almost no patent matter is that. The decision in front of you costs a fraction of it and happens years earlier.
2. **The capabilities that priced you out are now buyable a layer at a time.** Search, analysis, monitoring, evidence, claim mapping — at operating cost rather than institutional cost. Not that they have already diffused downward; that they can be, by anyone who goes and does it. Your competitor's legal budget was friction, not law, and friction is what collapsed.
3. **Your IP is the most underpriced asset on your cap table, and you are the one underpricing it.** Nobody is going to reprice it for you, and the window exists precisely because the evidence has not caught up yet.

If you are an investor, a large-company executive, or counsel advising them:

4. **Your model of the founder's cost structure is twenty years out of date.** The advice was correct arithmetic in 2006. Three of the four premises it rested on have expired, and the analysis was never re-derived — it was repeated.
5. **The vocabulary you are using was manufactured, and it is doing work you may not intend.** A founder enforcing a right built from their own product is not the behavior that term was coined to describe. It is being applied to them anyway, and the outcome gap that follows is measurable.
6. **The strongest evidence that this asset class is valuable is your own behavior.** Portfolios are being accumulated, licensing businesses run at software margins, and injunctions sought against smaller competitors — by the same firms whose advice to founders is that patents are not worth the trouble.

I will argue for all six with sources, and I will name every place the evidence is inconvenient — because it is worth knowing in advance how those places resolve.

They resolve in favor of the argument. Almost nobody owns this asset. Most holders abandon it. Nobody can reliably tell the good from the bad. Those are the facts usually produced to show that startup patents are not worth having, and every one of them is a description of **an asset trading below what it is worth**. Low ownership, no enthusiasm among the people who could bid, high abandonment, and a quality signal too noisy to price — that is not a case against the asset. It is the definition of a mispriced one.

Where something genuinely does not resolve — a contested figure, a study whose window closed before the thing I am arguing about began — I say so plainly and it does not move the conclusion. **I am not asking you to weigh this and decide. I am telling you where I think it lands, and showing you the work.**

The conversation I keep having

Several investors have now put a version of the same argument to me.

Patents are useful in theory, but they have limited practical value to a startup. If someone infringes one, you cannot afford to defend it. **A patent case costs three million dollars.** Therefore the patent is not much of a moat.

That figure comes up every time, always to two decimal places of confidence and never with a citation. Hold onto it. We will come back to it, because it turns out to be a real number describing a case that is almost certainly not yours.

Note that this is not only about hyperscalers. It applies to **any competitor** — the funded rival two blocks away, the systems integrator who rebuilt your architecture after a sales cycle, the vendor who read your documentation carefully. Size determines how much friction someone can impose on you. It does not determine whether they crossed your claims — the numbered sentences at the end of a patent that define exactly what it covers.

And notice what the argument actually says. It is not *your right is weak*. It is:

They have more money than you, therefore your right cannot be made economically consequential.

That is a claim about friction, not about law. Which is why this is a sequel rather than a new subject.

And it is worth naming the fear plainly, because it is the fear the whole argument runs on:

Your competitor's legal budget is not a moat to be feared. It was only ever friction — and friction is exactly what is collapsing.

A moat is a structural advantage. A legal budget is a purchased one, and what it purchases is your reluctance. That is a very different thing, and it prices very differently once the cost of finding out what is true stops being the barrier.

And I will put my own position up front rather than make you infer it: **I think intellectual property is the most underpriced asset on my cap table, and probably on yours.** Not because rights got stronger — they did not — but because the cost of operating one collapsed while the advice about them did not move. The rest of this article is the case for that, including the parts that cost money and the parts that can go wrong.

Nobody changed the law. The capabilities that priced founders out have come unbundled.

Start here, because everything else in this article is downstream of it.

The old asymmetry was real and I will not caricature it. Large companies have more capital, larger legal departments, greater distribution, more institutional memory, and more capacity to absorb delay. Those asymmetries still matter.

But people crossed those rights and got away with it — **not because the right was weak**, but because finding out what had happened, proving it, and forcing the issue cost more than it was worth.

The right existed the whole time. What protected the people crossing it was the cost of finding out.

That is the equilibrium now changing. Patent law did not become simple. Copyright, trademark and trade-secret law did not change. **What changed is the economics of understanding, monitoring, documenting, financing and exercising rights those laws already provide.**

If you take one thing from this article, take that. Everything below is evidence for it.

And note what it means for price. An asset whose value depends on being enforceable is worth what enforcement costs. When that cost was institutional, the discount was rational. **The discount survived. The cost did not.**

The number that scared you off your rights is real — and nobody repeating it has any reason to check it

Nobody who quoted me that figure could tell me where it came from. So I went and found it.

It is real, it comes from the American Intellectual Property Law Association — the professional body whose biennial economic survey is where almost every patent-cost figure in circulation originates — and it is **a single cell in one of its tables.**

In the *AIPLA Report of the Economic Survey 2017*, under Patent Infringement, All Varieties: median total cost for a case with **more than \$25 million at risk**, inclusive of pre-trial, trial, post-trial and appeal — **\$3.000 million.**

That is the cleanest origin, and nearby cells produce the same number in other editions. The 2015 survey shows \$3.0M for >\$25M at risk **through the end of discovery only.** The 2019 survey shows \$3.0M in the table for defending against a **non-practising entity** — a patent holder that does not make a product — at \$10–25M at risk.

The 2023 edition gives the full range: **approximately \$600,000 where less than \$1 million is at stake, to \$3.625 million where more than \$25 million is at stake.**

So the "\$3 million patent case" is the median total cost — **per patent**, both sides pooled — of a dispute with **more than twenty-five million dollars** at risk, all the way through appeal.

If that describes your dispute, you have a twenty-five-million-dollar problem and legal cost is not your binding constraint. If it does not, **the number was never about you**.

It has been quoted at founders for a decade as though it were the entry price of caring about a patent. It is the exit price of the largest cases in the survey — and it is quoted, almost always, by people who have never looked up which cell it sits in.

That is worth sitting with. A number nobody sourced has been setting founders' IP strategy. Not a legal doctrine, not a ruling, not anything anyone argued. A statistic about somebody else's twenty-five-million-dollar case, repeated until it became the reason not to try.

The category error

The "\$3 million patent case" prices the terminal branch. The founder's decision starts four gates earlier.



Figure 1 *The category error*

Here is the part worth stopping on, because it is not the part anyone argues about.

Nobody fabricated this. The number is real. It is just the price of a specific and very rare event — a full trial through appeal in a dispute with more than twenty-five million dollars on the table.

Almost no patent matter is that. So when you decided your patents were not worth pursuing, you were not making a bad decision about enforcement.

You were pricing the wrong event.

That is the first of three things holding the price down.

That is a different problem, and it has a different fix. A patent does not have to become cheap to *litigate* in order to become dramatically cheaper to *act on*. Almost everything in this article follows from that one distinction.

Almost no patent dispute ever reaches a courtroom

If the three million is the price of the last mile, the obvious next question is how many disputes ever travel it.

Lex Machina, the legal-analytics service that codes every federal docket, analyzed **10,128 patent cases terminated from 2022 through 2024** — excluding generic-drug cases, which follow their own statutory track:

Where 10,128 patent cases went

Non-ANDA patent cases terminated 2022–2024. Milestone counts, not mutually exclusive outcomes — a case reaching trial passed the earlier milestones too.

Milestone	Cases	Share
Terminated	10,128	100%
Reached a claim-construction hearing	1,094	10.8%
Reached summary judgment	576	5.7%
Reached trial	236	2.3%

Median time to termination: 208 days. Median time to trial, among cases reaching it: 1,040 days.

About 90 percent were, in the report's words, "resolved through settlements or procedural rulings rather than on substantive grounds" — 79 percent settlement, 11 percent procedural.

Source: Lex Machina Patent Litigation Report 2025. Expound Laboratories™ · Finality Assurance™

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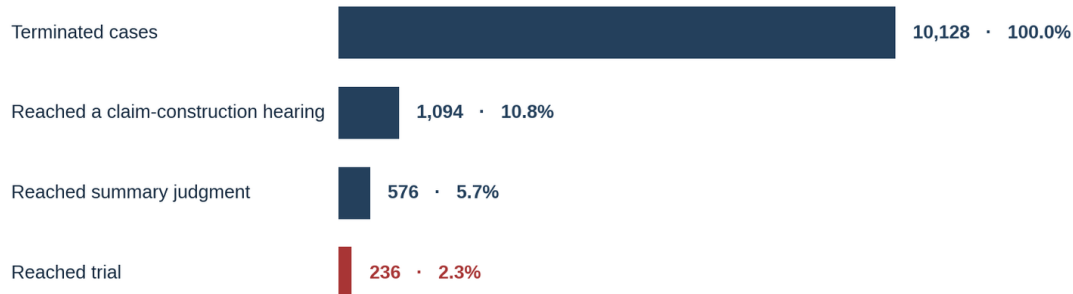
Figure 11 Where 10,128 patent cases went

Two of those milestones need translating. A **claim-construction hearing** is where a judge decides what the patent's words mean; it is usually the turning point of the case. **Summary judgment** is a ruling that the facts are not really in dispute, so no trial is needed.

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Where filed patent cases actually go

10,128 non-ANDA patent cases terminated 2022–2024. Milestone counts, not a mutually exclusive funnel — a case reaching trial passed the earlier milestones too.



Median time to termination: 208 days. Median time to trial, among cases reaching it: 1,040 days.

A courtroom is where these cases almost never go. Pricing every patent right at the cost of the rare one that does is the mistake — and the visible population still excludes every dispute resolved before a filing.

Bars are drawn to scale against the terminated-case population; no axis compression.
Source: Lex Machina Patent Litigation Report 2025. Expound Laboratories™ · Finality Assurance™
E^x

Figure 2 *Where filed patent cases actually go*

What this establishes: the overwhelming majority of patent disputes are decided by what happens *before* a courtroom — by what each side knows, can prove, and can afford to keep doing. Trial is the rare terminal branch, not the road.

Read 2.3 percent as a shape, not a constant. Two biases pull opposite ways: court data starts only at the complaint, so it never counts the investigations, notices, licenses and design-arounds that resolve quietly; and a three-year *termination* cohort censors out slow, trial-bound cases. Nor should anyone multiply that 2.3 percent by three million — cases that survive longer are selected cases, and costs cluster around the milestones needed to progress.

The precision is arguable. The shape is not. Nine cases in ten never reach a substantive ruling at all.

The decision that determines your outcome is almost never made at trial. It is made in the eighteen months before anyone files.

Which is the whole pricing question. If your right is worth what a trial costs, it is worth very little to you. **If it is worth what the eighteen months cost, that is a number you can now afford** — and it is not the number you have been quoted.

Intel invented the word that makes founders afraid to enforce their own patents

So the number is wrong and the path is wrong. Now the part I actually want to talk about, which is why so many people believe both anyway.

A founder who enforces a patent is called a troll. A corporation with fourteen thousand patents and a licensing division is called a sophisticated operator with a strong IP position. Same activity. Different vocabulary. And the vocabulary was **manufactured, by a large company, on purpose.**

The term came out of **Intel**, by way of then-vice president and assistant general counsel **Peter Detkin**.

Intel had publicly called an adversary a "patent extortionist" — contemporaneous reporting attributes that phrase to an Intel spokesman — and got sued for libel. In Detkin's own account, CEO Craig Barrett "didn't like having the issue played out in the press, and demanded I come up with a different name."

An internal contest produced the replacement. Who won it is disputed; Detkin credits his daughter's troll doll. The article that put the term into circulation is Brenda Sandburg's *"Trolling for Dollars"*, The Recorder, 30 July 2001. (Earlier bridge-troll uses exist; this is the one that stuck.)

A company being sued invented the word for people who sue it. That word is now the default frame through which a founder asserting a right is read.

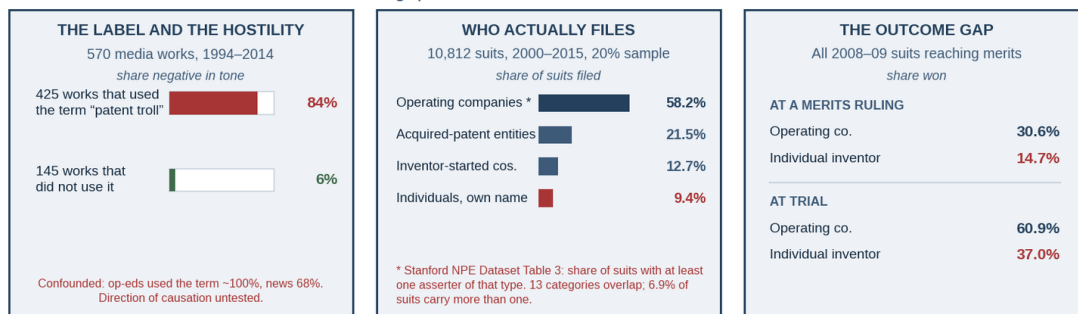
And there is measured evidence of the association. Edward Lee coded 570 **media** works on patent assertion between 1994 and 2014 — newspaper articles, op-eds, editorials and letters. Of the **425** that used "patent troll," **84 percent portrayed the entity negatively at least in part.** Of the **145 that did not use the term, only 8 — six percent — did.**

Before that does more work than it can bear: the comparison is confounded. The term appears in nearly every editorial and op-ed but only 68 percent of news articles, so the non-troll group is close to pure news — and opinion is negative by construction. Lee reads the association the other way round, and the experiment that would settle direction has not been run.

The label and the hostility travel together; which drives which is unproven. Fine. The correlation was never the interesting part. Who commissioned the word is.

The double standard, measured

A founder who enforces a patent is called a troll. A corporation with a licensing division is called a sophisticated operator. The label was manufactured. The outcome gap is real. What causes it is not established.



An individual inventor wins less than half as often as an operating company. Why is not established.

The gap narrows in ratio at trial and does not close. Pooled NPEs do converge with operating companies at trial — 60.0 against 60.9 — but individual inventors reach only 37.0. The authors report that entity-type differences at trial are not statistically significant in multiple regression, and say of the cases that settled without a ruling that they "do not even have a sense of which direction the bias, if any, would point." That a large share of the gap is procedural rather than substantive is a hypothesis this data does not test.

The vocabulary, though, was commissioned. "Patent troll" came out of a large company that had been sued for libel after calling an adversary a "patent extortionist," and whose CEO "demanded I come up with a different name." An internal contest produced it.

Sources: Lee, 19 Stan. Tech. L. Rev. 113 (2015); Miller et al., Stanford NPE Litigation Dataset, 21 Stan. Tech. L. Rev. 235 (2018); Allison, Lemley & Schwartz, 32 Berkeley Tech. L.J. 237 (2017); Wild, IAM, 22 Aug. 2008. Expound Laboratories™ · Finality Assurance™

Figure 8 The double standard, measured

So who actually files patent suits? The Stanford NPE Litigation Dataset codes a 20 percent random sample of suits filed 2000 through 2015 — **10,812 lawsuits** across thirteen assertor categories that deliberately overlap, since 750 suits (6.9 percent) involve more than one type.

The operating-company share moves depending on what you count, and all three of these are from the same paper:

- **58.2 percent** of those lawsuits include at least one product company as an assertor (Table 3).
- **53.2 percent** of *defendant-lawsuit pairs* do (Table 4) — the same litigation counted per defendant rather than per case.
- **About 45 percent** of pairs since 2010–11, down from roughly **70 percent** before 2006 (Figure 4).

None of them is wrong. They answer different questions, and the decline over time is the finding, not the pooled average. The authors also note that one further category — the IP subsidiaries of

product companies, another 1.9 percent — can reasonably be read as product-company litigation too, which would push the share up rather than down.

One figure does not move at all: individuals asserting in their own name — original inventors suing personally rather than through a company they own — are 9.4 percent of lawsuits and 11.3 percent of pairs.

The archetype the vocabulary was built around is under a tenth of the docket, and operating companies — firms that make and sell things — are the largest single category on any classification. The word was coined for a party that files roughly one suit in ten.

And the outcome disparity is real. Allison, Lemley and Schwartz examined **945 merits decisions** — rulings on who was actually right, as opposed to settlements or procedural dismissals — in patent cases filed in 2008–09. Definitive win rates: operating companies **30.6 percent**, universities 30.0 percent, failed startups 21.4 percent, **individual inventors 14.7 percent**, and patent-assertion entities — firms whose business is licensing patents rather than making things — 9.5 percent.

An individual inventor wins less than half as often as an operating company.

The gap narrows at trial without closing. Pooled non-practising entities — patent holders that do not make a product — converge with operating companies there — 60.0 against 60.9 percent, a difference of nothing. **Individual inventors reach 37.0 percent** against operating companies' 60.9. The ratio improves. The deficit remains.

And the authors' own caution travels with all of it: the entity-type differences at trial are "**not statistically significant when tested in a multiple regression,**" and as to the cases that settled without any ruling they write that they "*cannot confidently generalize*" and "**do not even have a sense of which direction the bias, if any, would point.**"

The disparity is established. Its cause is not — and I will say plainly which way I read it: most of that gap looks procedural and evidentiary rather than substantive, which is exactly the kind of gap operational capacity closes. This data does not test that, and I am not going to claim it does.

But notice where the burden actually sits. When someone cites 14.7 percent to argue startup patents are weak, they are treating an outcome as a property of the patent. **It is at least as likely to be a property of who could afford to litigate it** — and the person making the argument has no more evidence than I do.

The vocabulary was built by a defendant. The outcome gap is real but unexplained. Neither is evidence that your patent is weak — and both are routinely offered as if they were.

This is the second thing holding the price down. Not the law, and not the merits. **A word a defendant commissioned, and a statistic nobody has explained** — both doing the work of a valuation nobody has actually performed.

What those same companies do with the patents they own

The vocabulary is half the story. Here is the other half, and it is all on the public record.

They are accumulating. In calendar 2025, Samsung received **7,054 US patents** — more than two percent of every patent the office granted. Qualcomm took 3,749. Apple, 2,722. These are the annual IFI Claims rankings, published every January, and nobody in the industry treats them as a curiosity.

They are licensing. Qualcomm's licensing segment reported **\$5.58 billion** of revenue in its 2025 fiscal year at roughly a **72 percent pre-tax margin** — that is a patent portfolio operating as a business line, disclosed in a 10-K. Nokia's licensing unit reported **€1.5 billion**, with more than €800 million of contracted recurring revenue locked in through 2030. IBM booked **\$265 million** of intellectual-property income in 2025 — \$964 million if you include custom development income, which is a different business and is worth separating.

And they are asserting — offensively, against smaller parties. In 2022 Google affirmatively sued Sonos over seven patents. In October 2024 a jury found that Masimo's watches **willfully** infringed Apple design patents — *willfully* meaning knowingly, a finding that can multiply damages up to three times — and awarded Apple **\$250**, the statutory minimum. Apple's counsel told the court the ultimate purpose was an **injunction**: a court order to stop selling, worth far more than any damages award. (The injunction motion has been fully briefed since February 2025 and remains undecided, and the verdict reached Masimo's discontinued design, not its current one.)

Read that last one twice. **A company sued a smaller competitor over design patents, won a finding of willfulness, took two hundred and fifty dollars, and told the court plainly that the point was to stop the product.** That is intellectual property used as a competitive instrument. It is entirely lawful. It is also precisely the thing a founder gets called a troll for merely contemplating.

Now the advocacy. In 2017, Adobe, Amazon, Cisco, Dell, Google, Intel, Oracle and Salesforce formed the High Tech Inventors Alliance to press for measures that make patents harder to enforce. In 2025 and 2026, industry associations representing much the same membership publicly opposed the PERA and PREVAIL bills, both of which would strengthen patent holders.

I am not going to tell you what any of them intends. **I will put the two facts next to each other and let you do the arithmetic.** The companies whose associations argue that enforcement should be harder are, in several cases, the same companies accumulating thousands of patents a year, running licensing businesses at software margins, and suing smaller competitors to remove products from the market.

The strongest available evidence that intellectual property is valuable is how much of it the people telling you it is worthless are buying.

None of which requires pretending that extractive assertion does not exist. It does, the frustration behind the vocabulary is genuine, and anyone who has been on the receiving end of a shakedown will tell you so. But the vocabulary built to describe that behavior has been applied to a far wider class of people than it was ever meant to cover — including, routinely, founders enforcing rights over technology they actually built and ship.

The asymmetry is not that large companies enforce. It is that when they do it, it is called strategy.

What your investors decided in 2006 — the analysis never changed, and neither did the talk track

The vocabulary explains the reflex. It does not explain the advice. That has a date on it.

The canonical startup patent guidance is **Paul Graham's, from March 2006** — twenty years old:

"We do advise the companies we fund to apply for patents, but not so they can sue competitors... if they want to get bought, they should apply for patents because **patents are part of the mating dance with acquirers.**"

That is not bad advice. I want to be precise about this, because the easy move here is contempt and contempt is both unfair and unpersuasive. **That advice was correct when it was given.** It was the rational conclusion from four premises that were all true in 2006:

7. Finding out whether someone crossed your claims required expensive human hours.
8. Building the evidentiary record required expensive human hours.
9. Every route to acting on a right ran through an institution that billed by the hour.
10. Therefore the only two live uses of a patent were suing someone you could not afford to sue, and looking credible to an acquirer.

Given those four, *do not bother* is not lazy counsel. It is arithmetic.

Three of the four have expired. The fourth — the institution at the end of the road — has not, and I will be equally precise about that later. But the advice never got re-derived. It got repeated.

Twenty years on, the emphasis shows in the vocabulary. Across Y Combinator's Startup Library — index entries and the talk and podcast transcripts it carries — "moat" outnumbers "patent" roughly six to one: 207 against 33 raw, or about 112 against 19 once duplicates are removed, counted 22 August 2026.

That is a term count on one index page, not a corpus study, and the library is not silent: at least two items give direct patent and licensing guidance. What it shows is proportion. **Moats are a standing theme. Patent strategy is nobody's article.** In Sam Altman's *Startup Playbook* the words "patent," "IP," "intellectual property" and "moat" do not appear at all — he makes the moat argument without the vocabulary.

There are dissenting voices, and they are worth more than the silence.

Jason Lemkin, December 2017 — note that he concedes the market before he disagrees with it: *"99% of SaaS VCs don't care about patents in general, and even if they do, patent applications are not issued patents... But... Patents do matter, if you are going long... In both my start-ups, I made the core set of patent applications a First 90 Days priority."*

Steve Blank, in First Round Review, 4 October 2024, asking founders a question almost nobody asks: *"Do you have a patent strategy; pre and post-inoculation/defense? Can you go on the offensive and buy patents you think they may infringe?"*

And in life sciences the question was never open at all. **Which is the real line here — sector convention, not a pro- or anti-patent split.** Nobody re-derived the software convention after the cost structure underneath it moved.

You are not being given bad advice. You are being given 2006's advice, by people who have not been paid to check whether it still holds.

And that is the third. **A number nobody sourced, a word a defendant coined, and guidance written before the tooling existed.** Three suppressors, none of them a fact about your patent.

Note the actual line. It is not pro-patent versus anti-patent. It is **sector convention**, formed when operating a right was institutional work, and never re-derived after the operating cost changed.

I am not claiming investors are foolish. I am claiming their model was calibrated on a cost structure that no longer holds, and that nobody has been paid to recalibrate it.

Four things changed for founders. The advice changed none of them.

Here is what that advice was calibrated on, and what has moved since. It is not wrong so much as **stale**. It was formed when the operating costs below were three to five times higher and the tooling did not exist.

The fee structure changed in 2022 and almost nobody adjusted. The Unleashing American Innovators Act (Pub. L. 117-328 Div. W §107, signed 29 December 2022 and effective on enactment) raised the small-entity discount from 50 to **60 percent** (§107(a)) and the micro-entity discount from 75 to **80 percent** (§107(c)). A **small entity** is broadly a business under 500 employees; a **micro entity** is a small entity whose named inventors are under a gross-income cap and have few prior filings. Most seed-stage startups qualify as one or the other and pay the undiscounted rate anyway.

What it actually costs to file

Official USPTO fees only. A micro entity pays twenty cents on the dollar — a change made in 2022 that most founder-facing advice has not absorbed.

	Undiscounted	Small entity	Micro entity
Filing + search + examination	\$2,000	\$730	\$400
Full official-fee lifecycle *	~\$17,760	~\$7,034	~\$3,552
Track One prioritized examination	\$4,665	\$1,866	\$933

* My computation from the current USPTO fee schedule: filing, search and examination, plus the issue fee and all three maintenance fees. Excludes attorney fees, excess-claim fees, requests for continued examination, extensions, information-disclosure-statement fees and any foreign or international filing. Small-entity column assumes electronic filing.

Micro-entity gross-income limit: \$251,190. Track One buys a first action roughly seventeen months sooner.

Sources: USPTO fee schedule; Unleashing American Innovators Act of 2022, Pub. L. 117-328 Div. W §107. Expound Laboratories™ · Finality Assurance™
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Figure 10 *What it actually costs to file*

‡ The Track One request fee itself is \$4,515 / \$1,806 / \$903 under 37 CFR 1.17(c); the figures above add the processing fee 37 CFR 1.102(e) also requires.

† My own computation from the current USPTO fee schedule: filing, search and examination, plus the issue fee and all three maintenance fees. It excludes attorney fees, excess-claim fees, requests

for continued examination, extensions of time, information-disclosure-statement fees, and any foreign or international filing — which for most founders will exceed the official fees. The small-entity column assumes electronic filing, which is why it lands slightly below a flat 40 percent of the undiscounted figure: 35 U.S.C. §41(h)(3) cuts the basic filing fee a further 75 percent for small entities filing electronically. Paper costs more.

A micro entity pays twenty cents on the dollar. The gross-income limit for micro-entity status is **\$251,190** as of 9 September 2025 — and the separate higher-education basis has no income limit and no application-count limit, though small-entity status must still be certified and the institution itself cannot be the applicant. (Certify carefully. The statutory penalties — 35 U.S.C. §41(j) for small-entity assertions and §123(f) for micro-entity certifications — are not less than three times the amount the entity failed to pay. They date from the 2022 Act; the good-faith exception was added later, by Pub. L. 118-151 in December 2024. The USPTO announced it would begin assessing them on 12 June 2025, reaching back to fees paid on or after 29 December 2022.)

Track One is the most underused instrument available to a founder. For **\$933** at the micro rate, all in, USPTO figures show 1.7 months from *petition grant* to first action, against 20.8 months from *filing* on the regular path. Those clocks start in different places, so compare like with like: filing-to-petition-grant adds about 1.9 months, which puts the honest comparison at roughly **3.6 months to a first action against 20.8** — about **seventeen months sooner**, for under a thousand dollars. (Two caveats worth carrying. Track One also reports 5.0 months to "final disposition" against 29.3 on the regular path, but the USPTO counts a first *final rejection* as a final disposition, so that is not time-to-patent and I am not going to use it as though it were. And the Track One gauges are fiscal-year cumulative while the pendency figures are report-date and exclude cases with a request for continued examination.) For a startup whose fundraise, acquisition conversation or competitive window is measured in quarters, that is not a procedural nicety. It is the difference between an issued patent existing during the conversation and not.

And the tooling that used to arrive only as part of an institutional bundle — search, analytics, docket intelligence, claim checking — is now bought a layer at a time.

None of this makes a patent free. It moves the entry price from institutional to operational — and that is a different decision, taken by a different person, on a different budget.

Founders were left out of the Great Unbundling. That is the part that can change now.

Fees are the visible half. The larger change is in what you have to buy to operate a right at all.

Sophisticated IP capability used to arrive as one purchase: databases, patent search, legal research, technical specialists, claim analysis, drafting, docket monitoring, evidence management, discovery operations, institutional memory, litigation judgment, experts, capital. You bought most of it at professional rates, usually before knowing whether the matter deserved the expense.

That bundle is separating into layers you can buy, build, or skip.

The Great Unbundling

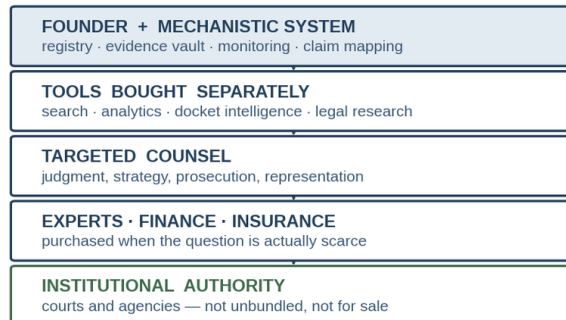
Sophisticated IP capability used to arrive as one purchase. It is separating into layers that can be bought, built, or skipped independently.

PREVIOUSLY BUNDLED



Bought as one unit, at professional rates, before you knew if the matter deserved it.

INCREASINGLY UNBUNDLED



AI does not eliminate the value of the great lawyer. It changes which hours of the great lawyer I need to buy.

Judgment, representation rights, privilege, advocacy and professional responsibility remain scarce — and adjudicative authority is not unbundling at all. What separated is the retrieval, organization and reconstruction that used to be sold at the same rate as the judgment.

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Figure 4 *The Great Unbundling*

And here is who got there first. Harvey is the largest of the legal-AI platforms — research, drafting, review, due diligence, litigation support. As displayed on its site on 22 August 2026 it reports more than 2,400 customers in over 70 countries, including **more than 75 of the AmLaw 100** — the hundred highest-grossing law firms in the United States — and more than 200,000 *professionals* using it — their word, and a deliberate one, since they sell beyond law. These are undated company-reported figures and they move fast: in March 2026 the same page said 1,000-plus customers in 60 countries, and Harvey's own funding announcement that month said 1,300.

Read that honestly: Harvey's core customers are **law firms**. More than seventy-five of the AmLaw 100 adopting AI is the incumbents capturing the efficiency first. The same goes for the 20 percent drafting-efficiency gain a large technology company's patent group reported from a test across its outside-counsel panel — an oral remark at a patent-law institute rather than a published study, and worth no more weight than that. Whatever it was, it accrued to outside counsel.

Which tells you the arbitrage is still open at the bottom. The capability is real enough that the most sophisticated buyers in the market have already bought it. It has not yet been priced into what a founder pays, or into what a founder is told their rights are worth. My claim is not that capability has already diffused downward. It is that **it is now purchasable, or buildable, below the firm layer** — by anyone who goes and does it before that stops being unusual.

What has not unbundled: judgment, experience, strategy, representation rights, privilege, professional responsibility, negotiation, advocacy, trial craft. I itemize the patent-specific version of this list later. And adjudicative authority is not unbundling at all.

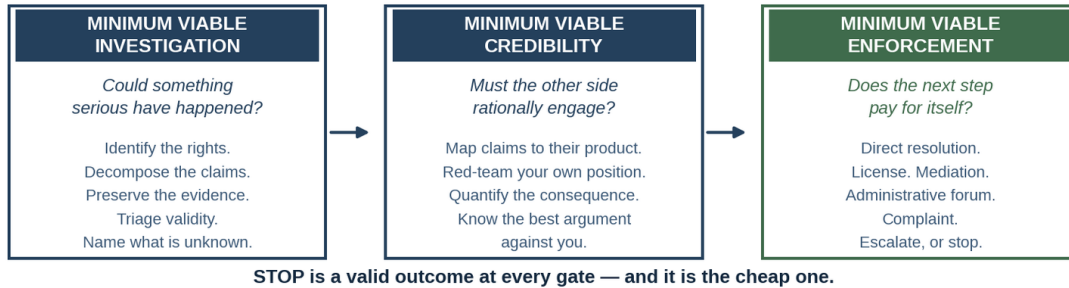
AI does not eliminate the value of the great lawyer. It changes which hours of the great lawyer I need to buy.

Three thresholds, not one

Unbundling only matters if it changes a decision. Here is the decision it changes.

Three thresholds, not one

Minimum Viable Enforcement was one gate in the predecessor article. Cheap investigation splits it into three, and each is a separate purchase decision.



The startup does not have to finance the terminal node before entering the decision tree. That is the whole of the argument. A right does not become cheap to litigate; it becomes cheap to find out whether litigating it was ever the question.

Minimum Viable Enforcement is a forward-looking economic hypothesis, not an observed market pricing result. Expound Laboratories™ · Finality Assurance™
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Figure 3 *Three thresholds, not one*

Minimum Viable Investigation. Could something serious have happened? Identify the rights, break the claims into their elements, examine the product you think is crossing them, preserve evidence, review the **prior art** — everything already public that could be used to argue your idea was not new — and name what you do not know.

Minimum Viable Credibility. Must the other side rationally engage? A credible rights holder does not send an angry letter. It has mapped its claims onto what the other side is actually doing, red-teamed its own validity, preserved evidence, and knows the strongest argument against it.

Minimum Viable Enforcement. Does the next escalation pay for itself?

You do not finance the terminal node to enter the decision tree. Stop is a valid outcome at every gate, and it is the cheap one.

Carrying the threshold forward, as published. *Who Can Afford Friction?* states Minimum Viable Enforcement as $E[B_E - C_E] > 0$ — benefit under each possible outcome *before* enforcement cost, against the cost under that same outcome. Where outcomes are alternatives rather than complements, their probabilities must be **jointly feasible and cannot sum above one.**

>

And the part a small company must not skip: **a positive expected value is not an affordable bet.** A \$10M payoff at 40 percent against a \$3M cost either way clears the threshold by a million dollars — and is a sixty percent chance of spending three million and recovering nothing. For a fourteen-person company that is not disappointing; it is terminal. **The binding constraint on a small holder is not the expectation, it is the variance they cannot absorb.**

Why the asset is cheap

Here is the part that should interest anyone holding equity.

Everything above describes a suppressed price. Now the mechanism — because four well-measured facts are usually presented as reasons not to bother, and every one of them is a description of an **underpriced asset class**.

Almost nobody owns it. The Berkeley Patent Survey — the standard study of what early-stage technology companies actually do about intellectual property, across 1,332 of them — found only **24 percent of software and Internet startups held any patent or application at all.** *That figure is from its population sample rather than all respondents, and the data was fielded in 2008 — before Bilski, before Alice, before the AIA.*

And those who could buy it, don't want it. The same survey found software firms rank patenting **last of the seven appropriability strategies it tested**, behind first-mover advantage, secrecy, complementary assets and the rest — reporting that patents "generally provide at best 'slight' incentives." Across venture-backed firms in every industry it covered, the survey found "in no category are patents reported to provide even a 'moderate' incentive."

Most holders abandon what they have. Moore's study of 1991-issued patents found **53.7 percent** of patentees let them expire early; Crouch's 2026 analysis puts the current figure nearer **60 percent** as maintenance fees have risen. Either way, most patents are dropped before full term.

And quality has been genuinely hard to tell apart. Across all 386 eligibility decisions from *Mayo* through 2023 in the **Federal Circuit** — the specialist appeals court that hears every patent appeal in the country, so its view is effectively the national one, **every judge deciding more than six of them found the patents invalid at least 73 percent of the time**; sixteen of nineteen did so 80 percent or more. **Section 101** is the provision governing what may be patented at all; it is the rule courts use to strike claims down as abstract ideas, and software claims live or die by it. *Note that study's own thesis is that §101 is more predictable than the usual telling — and where a district*

court had upheld validity, the Federal Circuit affirmed only 59.1 percent of the time — on just 22 such instances.

Read those four together and describe the market they make. **Low ownership. No enthusiasm among the people who could bid. High abandonment. And a quality signal so noisy that good and bad trade at the same price.**

That is not an argument against the asset. **That is the definition of one that is mispriced** — and I would rather own something in that condition than something everybody has already competed for.

What repricing it requires

An asset stays cheap while nobody can tell the good from the bad. So the question is what makes quality legible, and the answer is the same force this article has been tracking.

AI lowers the cost of prior-art search, invalidity analysis, claim construction, noninfringement mapping, §112 analysis — section 112 is the requirement that a patent describe its invention fully enough that a skilled engineer could build it — and design-around generation, **for the accused party too**. That symmetry is the whole point.

AI should not raise patent value uniformly. It should raise the *dispersion* in patent value.

Dispersion is the spread between the best and the worst. The average can sit exactly where it was while the gap between a strong right and a weak one widens — and that widening is the whole thesis.

Strong, supported, observable rights become easier to understand and activate. Weak, vague, unsupported ones become easier to expose and design around. **Count stops mattering. Quality starts being visible.**

And when quality becomes visible in an asset class priced at its historical average, the good end is worth more than the average and the bad end is worth less. Most advice still prices the whole distribution as one thing.

One honest boundary, because it is the boundary and not a hedge: none of this makes a patent the right instrument for every company. If your advantage is genuinely speed, distribution or a data asset, secrecy and execution may serve you better, and the Berkeley finding is partly *correct* rather than merely stale. The claim is not that everyone should file. It is that **the decision is being made on a price that no longer holds, by people who have not re-checked it.**

And one figure to stop repeating, in either direction: **"90 to 95 percent of patents are never licensed."** No primary study establishes it. It traces to a 2001 law review article whose author was estimating something narrower — the share of patents ever *litigated or licensed for a royalty*, as opposed to cross-licensed — and who wrote *"my sense"* and *"I suspect,"* adding in a footnote: *"I am troubled that it is only an estimate."* Later work restated it as a fact, widened it to licensing of any kind, and dropped the hedge. The well-sourced pairing is Lemley and Shapiro's, in *Probabilistic Patents* (2005): "Only 1.5 percent of all patents are ever litigated, and only 0.1 percent are litigated to trial."

An asset almost nobody owns, most holders abandon, and no one can price accurately is not a bad asset. It is a cheap one — and the thing that made it hard to price is the thing that just changed.

And it is not only an asset. It is an instrument.

Everything above prices intellectual property as something you hold. That undersells it, and it is the part my competitors have most consistently got wrong about me.

A right you can actually operate does work in the market **before** anyone files anything:

- **It sets the distance a competitor has to travel.** A well-layered claim set does not stop a rival — it makes going around you cost engineering quarters, and quarters are the currency they are actually short of.
- **It changes what your customers' procurement asks them.** An enterprise buyer's diligence question is not "is this good," it is "will using this create a problem." A rights holder who can answer that precisely is easier to buy from than one who cannot.
- **It moves the negotiation before the negotiation.** The party who knows exactly what they own, what it covers and what the evidence looks like is negotiating from a different place than the party who will have to go and find out.
- **It shapes the acquisition.** Not as the mating-dance signal Graham described, but as the thing a buyer's lawyers actually dig into — and where a clean **chain of title**, the unbroken paper trail proving the company and not a founder personally owns each right, plus a portfolio mapped to the product, is worth more than a bigger portfolio nobody can trace.

None of that requires a courtroom, and none of it shows up in litigation statistics. **The competitive value of a right is realised almost entirely in the shadow it casts**, and the shadow is proportional to how credibly you could act — not to how often you do.

Which is why underestimating a small company's IP capability is now a genuine strategic error rather than a safe assumption. **The question was never whether a founder could out-spend a**

large company's legal department. It is whether a founder can know their own position better than the other side knows theirs — and that is a question about operational discipline, which does not scale with headcount.

The repricing has not happened yet

Before I describe what I built, one thing has to be said about every number above — and it is the reason this is worth writing now rather than in 2030.

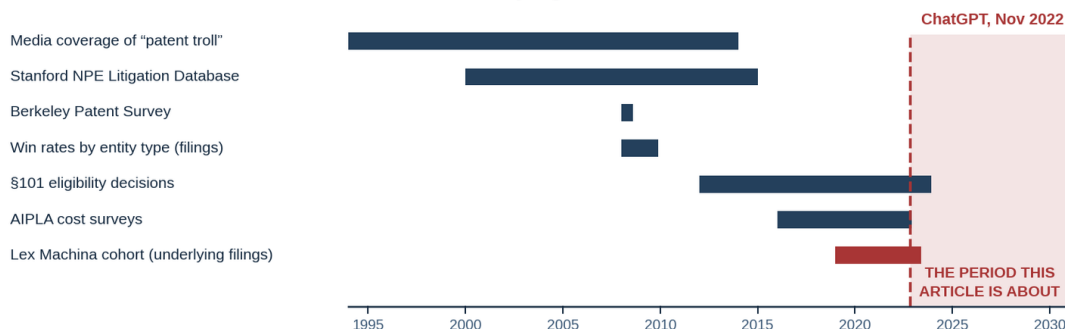
None of the data in this article can show you the repricing. Look at the vintages. A case terminating in 2024 was filed around 2023 if it moved fast, around 2021 if it went the distance. ChatGPT reached the public in November 2022. Harvey was founded that year and did not pass a thousand customers until 2026.

It is broader than one dataset. The win-rate study is 2008–09 filings. The Stanford dataset runs to 2015. The Berkeley survey was fielded in 2008. The media study ends in 2014. The eligibility analysis stops at 2023.

This is the one gap in the evidence I cannot close, and I want to be exact about why. It is not that better research exists and I did not find it. It is that the measurement period has not happened yet. No amount of diligence produces 2028 data in 2026, and anyone who tells you otherwise is extrapolating and calling it evidence.

Every number in this article is a pre-AI baseline

The evidence base for patent enforcement economics describes activity that concluded before the tooling existed. That is what makes the thesis testable rather than merely arguable.



A case terminating in 2024 was filed around 2023 if it moved fast, around 2021 if it went the distance.

So these are the right numbers and the wrong window. A baseline is exactly what you want before an intervention — but nobody should read them as evidence that anything has already changed. If the argument here is right, the 2028 and 2030 editions of these same reports diverge from them. If those editions look the same, the argument was wrong.

Windows shown are each source's own reporting period; the Lex Machina row shows the filings underlying its 2022–2024 termination cohort, inferred from its published 208-day median to termination and 1,040-day median to trial. Expound Laboratories™ · Finality Assurance™

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Figure 9 *Every number in this article is a pre-AI baseline*

Every number I have shown you is a pre-AI baseline — and that is not a weakness in the argument. It is the argument.

A baseline is what you want before an intervention. These are the best-measured facts the field has, and they describe the world as it was priced *before* the cost of operating a right collapsed. **If the repricing were already visible in the data, it would already be in the price.** The window exists precisely because the evidence has not caught up.

So do not read any of this as proof that AI has already changed enforcement economics. It cannot be. What it supports is a prediction, with named observables:

- **More disputes should resolve before a complaint is filed**, as investigation gets cheaper — shrinking the visible docket relative to actual assertion activity.
- **Small-entity filings and Track One usage should rise**, because the entry cost fell and the tooling arrived together.
- **The individual-inventor outcome gap should narrow** — if it really is procedural rather than substantive.
- **Patent-value dispersion should widen**: more weak rights exposed early, more strong rights activated.

And I want to be careful about the word I am using. This is not urgency. I am not telling you a window is closing. I am telling you that an asset is priced on an input that has already changed, and that repricing under those conditions is not a possibility — it is arithmetic waiting on measurement. The only open question is who is holding the asset when the measurement catches up.

If the 2028 and 2030 editions of these reports look like the ones I have cited, I was wrong — and I would rather stake the claim on observations that have not happened than overclaim the ones that have. But if you wait for those editions to be published, you will be buying the asset after it has been repriced.

What it looks like when a founder actually operates this way

So let me describe the destination first, because it is the part that should be obvious and it is the part nobody has drawn for you.

In a company that operates its intellectual property this way:

Nobody asks "do we own this?" and waits three weeks for an answer. The question resolves in minutes, because ownership, inventorship and the assignment trail are recorded facts rather than a research project someone has to run under deadline.

An invention is captured when it happens, not reconstructed when it matters. The dated package exists before anyone knows whether it will ever be worth anything — which is the only moment at which capturing it is cheap.

A competitor crossing your claims is something the system notices, not something you find out about at a conference. Monitoring is continuous and mechanical, and it produces a record rather than a suspicion.

Diligence is an export, not a scramble. When an acquirer's lawyers ask what you own and what it covers, you hand them a mapped portfolio instead of six weeks of your engineering team's time.

And the answer to "should we act on this?" has a computed value attached to it, stage by stage, rather than resting on how confident the founder felt that week.

None of that requires a litigation budget. None of it requires a law firm on retainer. Every item on that list is an operating capability, and operating capability is the thing that got cheap.

That is the world I am arguing is now available. Not a world where founders sue more. A world where founders *know what they have* — which turns out to be the part that was actually missing.

Give the thing a name: IP as code

Infrastructure used to be something you provisioned by hand, until somebody named the alternative *infrastructure as code* and the naming did real work: it turned a set of scattered practices into a destination people could point at and say **that is what we are trying to become.**

Intellectual property is where infrastructure was before the name existed.

So here is mine, and I hold it as a conviction rather than a slogan.

IP as code. Your rights are artifacts with versions, dependencies, tests and a build pipeline — not a folder of PDFs in a drive nobody opens between funding rounds.

What follows from treating them that way is not rhetorical. It is the same set of things that followed for infrastructure:

- **State is recorded, not remembered.** A registry is the system of record. Nobody's recollection is.
- **Changes have provenance.** Every claim traces to the disclosure and the commit that support it.
- **Checks run on every change,** not at the end, and not when someone remembers.
- **Promotion between stages is gated by a computed result,** not by whoever is in the room.
- **And the whole thing is reproducible** — the same inputs give the same answer next year, in front of an examiner or an opposing counsel, without me in the loop.

That is a category, not a product. Anyone can operate in it, with tools they buy rather than build. I am about to describe the version I built for myself, and I want the distinction to be clear before I do: **the model is the argument. My implementation is only evidence that the model is real.**

So I built it

That is the argument, and this is the evidence that I mean it: the version of the model I run against my own rights, every day, including against the claims in this article.

I do not trust an AI model to manage a growing intellectual-property portfolio.

A model can retrieve, compare, summarize, propose, draft and challenge. It can also omit, conflate, hallucinate, forget, and produce a confident answer that is unsupported. A preregistered evaluation of leading proprietary legal-research systems found hallucination rates of **17 percent** for one product, **about 19 percent** for a second and **33 percent** for a third — spring 2024 versions, methodology publicly disputed by the vendors, and reported here as a snapshot rather than a current state. What it establishes is the invariant:

MODEL_OUTPUT != AUTHORITY

So my architecture is inverted from the industry default:

The model is a worker inside the system. It is not the system of record.

The core is mechanistic automation I built. **The machinery is what makes the analysis reliable and trackable.** AI is a bounded contributor inside it.

The Four-Corners Test™, and why it is not the one you learned

During software development, my **Four-Corners Test™** decides whether a candidate innovation has earned its way into the registry at all.

Patent law already has a four-corners idea, and mine is named after it on purpose. But it is not the same test, and the difference is the whole reason I built my own.

The legal doctrine asks whether a filed **specification** — the written description you file with a patent application, explaining what the invention is and how it works — read within its own four corners, shows that the inventor **possessed** the invention claimed. *Possession* is the legal test: not that you wanted the result, but that you had the thing. In *Ariad Pharmaceuticals v. Eli Lilly*, 598 F.3d 1336, 1351 (Fed. Cir. 2010) (en banc), the Federal Circuit put it this way: the test "requires an objective inquiry into the four corners of the specification from the perspective of a person of ordinary skill in the art."

That is a sound legal test. It has three properties that make it useless as an operating control.

It runs **once, and late** — on a document already filed, in an **office action** or a courtroom, months or years after the work happened. An office action is the examiner's written rejection; the years of back-and-forth answering them is called **prosecution**, and it is where most of the real money and most of the permanent damage happen. It runs **through a constructed reader** — the hypothetical person of ordinary skill, who is *permitted to supply background the document does not contain*, so whether you passed turns on what that imagined reader is deemed to have known. And it is

argued, not computed — the output is a holding after briefing, not something you can run on a Tuesday afternoon against forty candidates.

Mine keeps the boundary and changes all three.

It asks whether the conclusion drawn about a candidate can be **reconstructed from the bounded package and the authoritative evidence that package cites, with hidden context empty**. Load-bearing dependence on an unrecorded conversation, on private memory, on an uncited source, on an unavailable path, on an unstated assumption, or on a limitation that quietly vanished during summarization — any one of those fails the package.

What I have just described is the specification, not the implementation. The exact mechanisms that carry out the Four-Corners Test™ and compute a Finality Verdict are **trade secrets and are not described in this article**, and patent applications covering both are **pending**. Nothing above is a disclosure of implementation, and none of it should be read as one.

That boundary is not coyness, and it is worth explaining because it is the same decision this article has been telling you to make deliberately rather than by accident.

Disclosure cuts in both directions at once. Publish a mechanism and it stops being a trade secret the moment you publish it — that is definitional, not a risk. Describe a mechanism in a specification and decline to claim it, and under *Johnson & Johnston* you have dedicated it to the public — a rule I come back to in detail below. So there is a line, it has to be drawn on purpose, and where I have drawn it is this: **the property is public, the mechanism is not**. You can hold me to what the test requires and what fails it, which is the part that matters to a reader. How it is carried out is the part I am keeping.

If you take one operational habit from this section, take that one — decide what you are publishing and what you are keeping *before* you write the sentence, not after somebody asks about it.

Reconstructability is a property of the package, not of the reader.

That single move is what makes mine the stricter of the two. The legal standard lets a skilled reader close gaps with what he is presumed to know. Mine lets nobody close anything. A package that passes my test would pass the legal one. The reverse does not hold — which is exactly the asymmetry I want, because the legal one is the one that gets applied to me by someone else, on their schedule.

And mine runs where the technical work happens, on every candidate, mechanically, with a recorded result.

What it actually buys me.

The failure the MPEP warns about at §2161.01 — claims that "define the invention in functional language specifying a desired result" without describing how the result is achieved — is a four-corners failure by another name. Mine catches it at conception, when the fix costs an afternoon of writing down the mechanism. The legal one catches it at examination or in litigation, by which time §132(a) may have closed the repair: you cannot add new matter to cure a disclosure that never described how.

It takes my own memory out of the evidentiary chain. Conventional practice reconstructs an invention months later from interviews and recollection — and recollection is precisely what a deposition is built to break. A package that stands on its own has nothing there to break.

And it forces the distinction founders collapse most often: **having had an idea** versus **being able to show possession of a mechanism**. Only the second is patentable. Only the second survives contact with an examiner.

One boundary matters here, and I hold it deliberately: **a Four-Corners pass is a sufficiency test, never a verdict**. It establishes that the package is complete enough to be judged. It does not say the candidate is ready to advance. That decision belongs to something else, and I do not get to make it either.

The registry and the engine

A **patent registry** — the system of record for everything I might one day file on — is maintained mechanically rather than by hand: candidate identity and status, provenance, prior-art findings, technical artifacts, and links to the code, benchmarks and deployment evidence that exist for each candidate. *Provenance* here means the recorded chain of where each piece of evidence came from and when.

That registry feeds a **patent engine** that assembles filing materials, traces supporting evidence back to its source, exposes missing information, and flags what requires a human to resolve.

The Finality Verdict: the part I am not allowed to override

Every stage of that pipeline runs under a Finality Verdict.

That is not a metaphor. It is the mechanism my technical papers describe, turned on my own IP operations.

A Finality Verdict is a **present-tense answer to a narrow question**: given the evidence that exists right now, may this candidate advance to the next stage for this specific purpose? Not "is this a good invention." Not "am I confident." One claim, one purpose, one moment, decided on evidence.

Here is how it is produced. *What follows are its properties. How they are computed remains a trade secret, as above.*

Before any work starts, each stage declares an **obligation roster** — the list of things that must be true before a candidate is allowed through. Declaring it in advance is the point: I cannot discover after the fact that the checks I happened to run were the checks that mattered.

Evidence is then gathered against that roster and **typed by what it is capable of establishing** — a direct source can settle a question outright, a secondhand one cannot, and an assertion with nothing behind it settles nothing at all. Each type carries a ceiling on how good the answer it supports is permitted to look.

The verdict is then **computed, and written by nobody**. It is the *meet* — the mathematical term for the worst value in a set — taken over the whole roster, on a four-value scale:

BLOCKED < UNKNOWN < DEGRADED < VERIFIED

One blocked obligation blocks the stage regardless of how strong everything else is. **Bad news wins.**

There is no interface through which I can mark a candidate ready. If the disclosure does not support a limitation, the stage returns BLOCKED and names the obligation that failed. If the prior-art sweep has not run against the current claim set, the honest value is UNKNOWN — an absence recorded as an absence, never quietly rounded up to a clean pass.

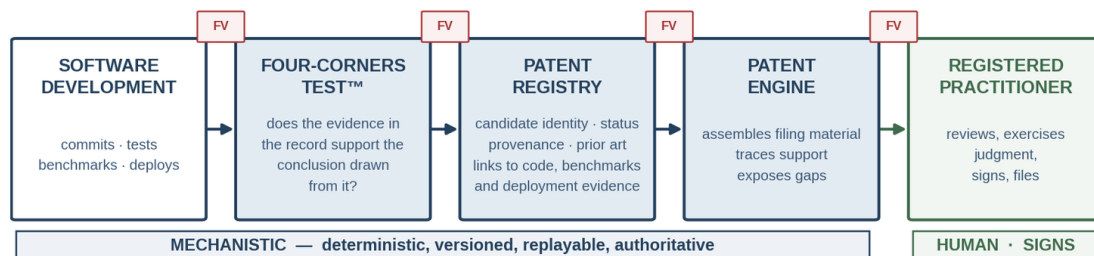
That last distinction is the one that matters most in practice, and it is the one ordinary process gets wrong. In most workflows, "*we checked and it was fine*" and "*nobody checked*" look identical from the outside. They produce the same green tick. Here they cannot: one is VERIFIED and one is UNKNOWN, and UNKNOWN does not let the stage proceed.

And a verdict expires. Because the system runs under **Continuous Finality™** — the property that a verdict is valid *at the moment of reliance* rather than forever — a change in the underlying basis reopens it. New prior art lands, a claim is amended, an **embodiment** — one specific worked version of the invention, described in the patent as an example of it — is deprecated in the code:

the affected set is recomputed from recorded dependencies, the earlier verdict narrows or withdraws, and the historical record stands as a true account of what was known then. **A patent decision that was correct in March does not silently stay green in August merely because nothing re-examined it.**

IP as code: the pipeline

The registry is the system of record. The model is a worker inside it. Nothing reaches a filing without a registered practitioner reviewing and signing.



FV = a Finality Verdict gates every stage transition. Each stage declares its obligation roster in advance; the verdict is computed from typed evidence as the meet over that roster, and is settable by nobody:

BLOCKED < UNKNOWN < DEGRADED < VERIFIED One blocked obligation blocks the stage. Bad news wins.

Continuous Finality™: when a basis changes — new prior art, an amended claim, a deprecated embodiment — the affected set recomputes, the earlier verdict narrows or withdraws, and the historical record stands as what was known then. AI operates inside the band, bounded: retrieval, alternative language, comparison, drafting assistance, adversarial review. It never becomes the record.

Source code is never required for a patent — §112 requires an algorithm (Aristocrat v. IGT; MPEP §2161.01).

The gain is evidentiary: it reduces the risk that the algorithm described is not the algorithm built.

Under 37 CFR 11.18 a natural person must review and sign; an AI tool cannot. Expound Laboratories™ · Finality Assurance™

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Figure 5 IP as code — the pipeline

What the verdict buys me.

It removes me as a point of failure. The founder-run IP process fails in one characteristic way: the founder, who is busy and who wants the filing to go out, decides it is good enough. I built the one system where that decision is not available to me. Not discouraged — unavailable.

It converts optimism into a recorded state. Every soft spot in a candidate has a name, a stage and a value, sitting in the registry. I can tell you today which of my candidates are VERIFIED and which are DEGRADED and why, without opening a single file or remembering a single conversation.

It makes the record admissible-shaped rather than anecdotal. What was known, when it was known, and on what evidence — computed and timestamped as work happened, not reconstructed afterward by the person with the most at stake in the answer.

And it keeps me honest. Every factual claim in this article was registered against the same roster and graded by the same rules — including the ones that rest on nothing but my own word, which are typed as assertions and cannot be graded higher no matter who made them. **A gate whose result I would only trust when it is favorable is not a gate**, and one I can talk my way past for my own claims is worth even less.

This is the unlock in time, cost and risk. Patent work has traditionally meant reconstructing an invention months later from interviews and recollection — losing embodiments, obscuring chronology, and opening gaps between what was described and what was built. My process captures it where the technical work happens, and refuses to let a stage pass on assertion.

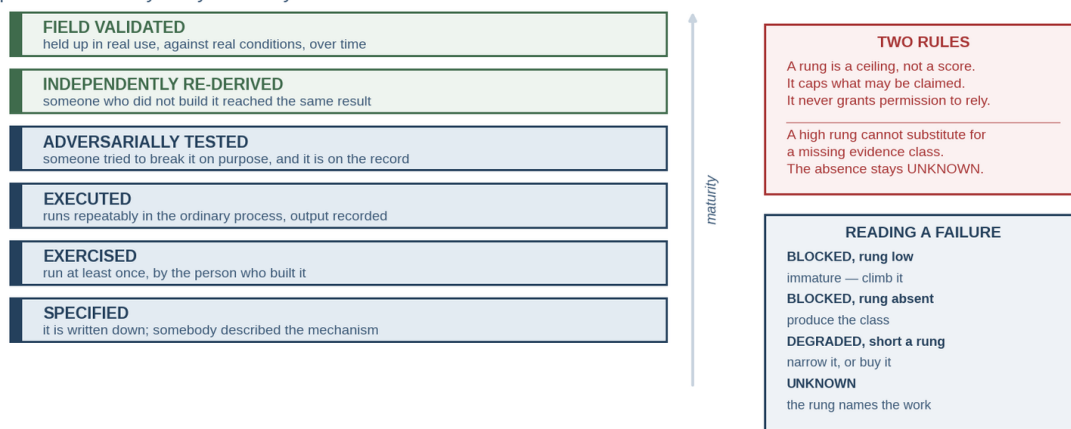
The Confidence Ladder™: what to do when the verdict is not VERIFIED

A verdict tells you whether you may proceed. It does not tell you what to do when you may not, and "not yet" is the answer you will get most of the time. That gap is where founders stall, and it is what the **Confidence Ladder™** is for.

The ladder types **evidence maturity** — how far a piece of evidence has actually been taken — on six rungs:

The Confidence Ladder™

Evidence maturity, typed. A rung caps what may be claimed — it never grants permission to rely. Only a Finality Verdict does that.



The verdict is the gate. The ladder is the map of what it would take to open it.

It converts "I am not sure" into "I am on this rung, the decision needs that rung, and moving up costs this" — which is a purchase decision rather than an anxiety.

Most claims in the accompanying article sit at executed; a few at adversarially tested. Nothing sits at independently re-derived or field validated. Expound Laboratories™ · Finality Assurance™

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Figure 12 *The Confidence Ladder™*

Two rules govern the ladder, and they are the entire discipline.

A rung is a ceiling, not a score. It caps how strongly a claim may be stated and how much scrutiny it attracts. **It never grants permission to rely.** Only a Finality Verdict does that, and no amount of climbing raises one. Confidence and permission are different objects, and collapsing them is the single most common way a founder talks themselves into a filing.

A high rung cannot substitute for a missing evidence class. If the roster requires a prior-art sweep and no sweep has run, nothing else fills that hole — not a strong benchmark, not a confident engineer, not a good week. The absence stays UNKNOWN. This is the rule that stops confidence from being laundered into completeness.

Where it earns its keep: reading a failure

When a stage does not pass, the verdict tells you *that* it did not. The ladder tells you *why*, and the two together tell you what the cheapest next move is. Four shapes cover almost everything:

- **BLOCKED, and the failing obligation sits low on the ladder.** The evidence exists; it is just immature. The fix is work you already know how to do — run the thing, record the output, climb a rung. This is the good failure, and it is the most common one.
- **BLOCKED, and the obligation is absent entirely.** No rung applies, because there is nothing on the ladder. The fix is not more confidence — it is producing that class of evidence at all. Different budget, different decision, and worth recognizing as different before you spend a month on the wrong one.
- **DEGRADED,** because the best evidence sits at *executed* and the claim needs *adversarially tested*. Now you have a genuine choice, and the ladder is what makes it visible: **narrow the claim to what *executed* actually supports, or buy the adversarial review.** Both are legitimate. Guessing between them is not, and guessing is what happens when nobody has typed the evidence.
- **UNKNOWN.** You do not know yet. The ladder tells you exactly which rung you would have to reach to convert it into an answer — which turns an anxiety into a quotable piece of work.

The verdict is the gate. The ladder is the map of what it would take to open it.

That is the whole value for a founder, and it does not require my machinery: **it converts "I am not sure" into "I am on this rung, the decision needs that rung, and moving up costs this."** That is a purchase decision, which is the same frame as the three thresholds earlier in this article — and a purchase decision is something you can actually make.

Applied to this article, since I should eat this too. Most claims here sit at **executed**: figures read from primary sources, then read again by a second verification lane that did not see the first one's answer. A handful sit at **adversarially tested**, because a hostile review was run against them and its findings are recorded. **Nothing here sits at independently re-derived or field validated**, and I am not going to imply otherwise by staying quiet about the top two rungs.

What working against the code buys — stated precisely

The obvious version of this claim is legally wrong, so let me be exact.

Source code is never required for a patent. *Aristocrat Technologies v. International Game Technology*, 521 F.3d 1328, 1338 (Fed. Cir. 2008), says an applicant "was not required to produce a listing of source code" — and says it while **invalidating** claims that disclosed no algorithm at all. That is the point. The case is a warning, not a safe harbor. What is required, where a computer-implemented claim is written in functional terms, is an **algorithm** — expressible, in the MPEP's words drawing on *Finisar v. DirecTV Group*, "as a mathematical formula, in prose, or as a flow chart." Deployment is not required for patentability either.

The benefit is **evidentiary and operational, not statutory** — and it is real. The Manual of Patent Examining Procedure — the rulebook examiners actually work from — warns at §2161.01 about claims that "define the invention in functional language specifying a desired result" where the specification does not describe how the function is performed, and is explicit that "it is not enough that one skilled in the art could write a program to achieve the claimed function." Working from the implementation attacks that gap at its root: **it reduces the risk that the algorithm described is not the algorithm built.**

There is a cost running the other way, and I run into it directly. Code-derived language is full of "module," "engine," "handler," "service" — and **"module" is a classic nonce word** — a word that names nothing in particular. Get the direction of the rule right, because it is easy to state backwards: writing "means for" triggers **§112(f)** automatically, while a nonce word does not. The presumption still runs *against* §112(f). What *Williamson v. Citrix Online*, 792 F.3d 1339 (Fed. Cir. 2015) (*en banc* — the full court sat, so the holding binds hardest) did was make that presumption **easy to rebut**: it overruled the line of cases calling the presumption "strong," so an accused infringer need only show by a preponderance that the term recites a function without sufficient structure. The consequence if they succeed: a claim written that way shrinks to only the structures your specification actually described, plus close equivalents of them. You meant to claim a category and you claimed your own implementation. Drafting from an implementation makes that *more* likely, not less. The pipeline flags nonce terms; deciding what to do about one is judgment.

Claim layering and design-around closure — bounded

Here is what I actually claim, and what I do not.

Mechanizable, because it is formally checkable:

- **Dependent-claim structure.** 35 U.S.C. §112(d) requires a dependent claim to reference a prior claim and specify a further limitation; the dependent claim is **construed to incorporate** the parent's limitations by reference. A dependent claim that does not narrow can be **invalidated** — *Pfizer v. Ranbaxy*, 457 F.3d 1284 (Fed. Cir. 2006), where claim 6 was held invalid under §112 ¶4 because it was "completely outside the scope" of the claim it referenced.
- **Multi-agent claim steps**, flagged for review.
- **Claim-type coverage** across the statutory categories — which are **process, machine, manufacture and composition of matter**. "System" and "computer-readable medium" are drafting conventions mapping onto machine and manufacture, not categories.
- **Claim-limitation-to-disclosure traceability**, and nonce-term detection.
- **Deadline, priority, marking and maintenance control** — mechanical rather than remembered.

Let me kill an overclaim before someone else does. Automated dependency and antecedent-basis checking is **commodity software** and has been for years — ClaimMaster, Patent Bots, PatentOptimizer, Rowan Patents. USPTO Patent Center itself emits improper-dependency warnings on DOCX filings — warnings only, non-blocking, and they never reach the examiner. I have not invented this category. What I do differently is run these checks **as a gate on every draft rather than as a final pass**, bound to a governed registry and to running code, with a computed verdict rather than a reviewer's recollection.

Not mechanizable, and I buy it:

Claim construction risk. Prosecution disclaimer pricing — an argument you make to distinguish your claim from prior art *can* narrow that claim permanently, where it amounts to a clear and unmistakable disavowal of scope, and after *Aylus Networks v. Apple*, 856 F.3d 1353 (Fed. Cir. 2017), even a preliminary response in an **inter partes review** (an IPR: a cheaper USPTO proceeding where a challenger asks the agency to cancel your patent) can do it, before the review is even instituted. That is a permanent forward cost no automation aimed at getting a patent allowed can price. Examiner interviews. Appeal versus **RCE** — a request for continued examination, which buys another round with the examiner instead of going to the appeal board. Portfolio-level sequencing, where *In re Cellect* and its narrowing in *Allergan v. MSN*, 111 F.4th 1358 (Fed. Cir. 2024), make **double patenting** (two of your own patents claiming things too close together, which can kill the later one) and **term adjustment** (extra patent life granted when the USPTO itself was slow) a cross-application, multi-year problem. Whether to assert at all.

What mechanization closes — and what it does not

The honest division. The left column is formally checkable.
The right column is judgment, and no pipeline of mine touches it.

MECHANICALLY CHECKABLE	REQUIRES HUMAN JUDGMENT
Dependent-claim well-formedness every limitation of the parent, plus one more — a set-containment test	Claim construction risk how a court will read the words you chose
Single-actor closure does one party perform every step, or is the claim divided-infringement bait?	Prosecution disclaimer pricing every argument distinguishing art is a permanent enforcement cost
Statutory-category coverage method, system and non-transitory medium claims all present	Examiner interviews and appeal strategy appeal, or file an RCE? Judgment, and the reversal rate is not stable
Claim-to-specification traceability every limitation traced to disclosure that supports it	Portfolio-level sequencing obviousness-type double patenting and term adjustment across a family
Disclosure completeness alternative embodiments enumerated from the code before filing	Whether to assert at all the strongest argument on the other side, priced honestly
Deadline and status control priority, maintenance, marking, renewal — mechanical, not remembered	Representation and signature a registered practitioner signs. A tool cannot, and may not.

I claim the left column. I buy the right one.

Design-arounds are closed at disclosure time or not at all — 35 U.S.C. §132 forbids new matter, and a continuation is limited to what the original specification supports. Expound Laboratories™ · Finality Assurance™
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Figure 6 What mechanization closes, and what it does not

And a limit on design-around closure that is easy to miss. Closure happens at disclosure time or not at all — §132(a) bars new matter — you may not add anything to an application that was not in it when filed — and §120 limits a **continuation** (a follow-on application that claims more of what the original already described) to what the parent disclosure supports under §112(a). *Liebel-Flarsheim v. Medrad* is the standing warning: Liebel won the broad claim and then could not enable it — *"beware of what one asks for."*

But enumerating from the implementation enumerates **what I built** — and a design-around is by definition **what I did not build**.

Here is what changed about that, and it is the part I did not expect.

Closing design-arounds used to be gated on breadth of engineering expertise, not on effort. To enumerate the ways someone could achieve your result without practicing your claim, you had to *know those ways* — across disciplines you had no reason to have studied. A software founder could enumerate software alternatives and stopped there, because that was the edge of what they could imagine competently.

That expertise gap has largely closed. It is now genuinely tractable to work outward from a mechanism and enumerate the places it could be realized instead: in a different software

architecture, yes, but also in firmware, in a hardware pipeline, burned into an ASIC as fixed logic, pushed into a network layer, moved into a compiler pass, or split across a boundary so no single party performs every step. Those were not hypothetical closures before. They were simply out of reach without hiring the discipline expertise to see them, at a cost that put the exercise out of range for anyone but a large company.

Two things follow, and the second matters more than the first.

The obvious one is defensive. A claim set drafted by someone who could only imagine software alternatives is a claim set with hardware-shaped holes in it. Now you can find those holes before filing, when §132(a) still permits you to do something about them, rather than after issue, when it does not.

The one that is worth more is not about IP at all. The same exercise — *where else could this mechanism be realized, and for whom* — enumerates applications of your technology you had no route to imagining. It surfaces use cases in adjacent industries, deployment contexts you have never sold into, and problems your mechanism solves that you did not know were problems. I have found more strategically useful information in that output than in the claim-drafting it was meant to support.

And to name the objection directly: this is not cheating. Nobody says a company with a hardware group and a compiler group and a network group is cheating when it drafts claims covering all three. It has simply been able to afford that view of its own invention. What changed is not the legitimacy of the exercise — it is who can run it. That is the same argument as everything else in this article, arriving in a place I did not expect to find it.

Worse, exhaustiveness is not free. Under *Johnson & Johnston v. R.E. Service*, 285 F.3d 1046, 1054 (Fed. Cir. 2002) (en banc), "when a patent drafter discloses but declines to claim subject matter... this action **dedicates that unclaimed subject matter to the public**" — gone, unavailable under the **doctrine of equivalents**, the rule that otherwise lets you win against a rival whose trivial change dodges your claim's literal wording.

The rule is bounded. Later cases require the disclosure be specific enough to identify what was disclosed and not claimed, and be presented as an alternative to a claim limitation. It does not spring on every stray sentence. **It springs on exactly the kind of systematic embodiment enumeration I was describing. Disclosing everything and claiming less narrows coverage rather than widening it.**

So the claim has to be narrow: the pipeline reduces the *forgotten-variant* failure mode, and does nothing about the adversarial-imagination problem. Deciding what to disclose and what to claim is judgment, and I do not automate it.

Three hard boundaries

A registered practitioner signs. A *registered practitioner* is a patent attorney or patent agent licensed by the USPTO to act before it — a credential distinct from a general law license. Under 37 CFR 11.5(b)(1), drafting a specification or claims *is* practice before the Office, so it is work only such a person may do for anyone but themselves. The USPTO's April 2024 AI guidance permits these tools and requires no general disclosure of their use — but "simply relying on the accuracy of an AI tool is not a reasonable inquiry," an AI tool may not sign, may not hold a USPTO.gov account, and is not a "user" of the filing systems.

A human must have conceived it. An inventor must be a natural person — *Thaler v. Vidal*, 43 F.4th 1207 (Fed. Cir. 2022) — and the USPTO's current guidance (90 FR 54636, 28 November 2025, expressly rescinding the February 2024 guidance) holds that the same inventorship standard applies regardless of AI use. Claims the pipeline proposes are reviewed for human conception before they are claimed. Where I cannot identify a human conceptual contribution, it is not filed.

I am not selling you this, and that is the point

This runs on my own rights only — internal infrastructure, not a service. Let me be flatter about that than a founder usually is, because the reflex here is to hint that a product is coming.

I have no intention of productizing the patent machinery. Not as a SaaS, not as a licensed tool, not as a consulting practice. If I sold it, every competitor who bought it would arrive at the same position I hold, and the advantage would price itself away on contact. **The advantage exists precisely because it is not for sale.** That is also why the mechanisms are trade secrets rather than a blog post: a capability you can buy is a cost line, and a capability your competitor cannot buy is a moat — the real kind, not the purchased kind I told you not to fear.

Which brings up the part that matters more than the patent pipeline itself.

The Four-Corners Test™, the Finality Verdict and the machinery underneath them are not patent tools. They are general-purpose governance components that happen to be pointed at my IP operations in this article. The same components run inside other things I have built — including **Governed Multi-Route Selection and Constrained Policy Reinforcement Learning** — GMRS and CP-RL, the governed model router — where the question is not *may this patent candidate advance* but *may this route be taken, on this evidence, for this purpose*. Same roster, same typed evidence, same meet, same refusal to let anything pass on assertion. Neither has been released, no release date is committed, and plans change.

That is what a portfolio actually is, and it is what gets underpriced. Investors count filings. What matters is whether one mechanism serves several products, because that is the difference

between an asset that is worth its filing cost and one that compounds across everything you build next. A patent covering a component I use in three places is not three times as valuable as a patent covering a feature. It is a different category of thing.

If you take nothing else operational from this section: **when you are deciding what to protect, protect the mechanism, not the feature.** The feature ships once. The mechanism keeps showing up.

For anything requiring practice before the Office, or representation of any party including my own entity, I retain a registered practitioner. Note the entity trap, because it catches founders: under 37 CFR 1.31(a)(1) — a rule that already required this of juristic-entity applicants, and which was extended effective 20 July 2026 to patent owners and to foreign-domiciled applicants — **a juristic entity must be represented by a patent practitioner** — *juristic entity* meaning a company rather than a human being — and 28 U.S.C. §1654 permits an individual to conduct *his own* case — the rule that a corporation or LLC must appear through counsel comes from the case law construing it, *Rowland v. California Men's Colony*, 506 U.S. 194 (1993). If your rights sit in the company, counsel is not optional. Check who the **applicant of record** is: the party the USPTO believes owns the application, which is not always the party you assume.

Continuous monitoring, and what your records are actually for

The system does not stop when an application is prepared. I continuously scan for relevant prior art and automate the tracking of possible violations. The objective is not to ask a model whether someone stole an idea. It is to build a chain that survives inspection:

Right → protected element → what they are actually doing → evidence → provenance
→ uncertainty → next lawful step

What I am actually watching for. Monitoring is only useful if it names specific signals in advance rather than looking for "something suspicious," so here is the shape of mine — the categories, not the thresholds:

- **My marks, appearing where they should not.** *Expound* and the marked terms — Finality Assurance™, Continuous Finality™, the Four-Corners Test™, the Confidence Ladder™ — in product copy, documentation, conference material, domain registrations and app-store listings. Trademark rights erode through unpoliced use, so this one is not optional.
- **My unmarked technical names, appearing as someone else's.** Accepted Work per Dollar, Finality Verdict, Finality Account, GMRS, CP-RL and the rest are descriptive names I do not

claim as marks — but a competitor adopting the vocabulary wholesale is a signal about where they got it, whatever it does or does not establish legally.

- **The distinctive shape of the mechanism, described in someone else's words.** A verdict computed as the worst value over a roster declared in advance. An evidence type that caps a claim rather than scoring it. A gate with no override. Nobody has to use my vocabulary to have taken the structure, and the structure is what the claims cover.
- **Copyright: my text and figures, reused.** Article passages, diagrams, the technical papers, and code. This is the cheapest category to prove and the cheapest to enforce, and founders routinely ignore it because it feels less serious than patents.
- **Prior art landing against my own pending claims** — which is the same monitoring loop pointed at myself, and the reason a verdict withdraws rather than standing.

What comes out is not an accusation. It is a dated, sourced observation with its uncertainty attached, sitting in the registry until there is enough of it to justify the next lawful step — or until it goes stale and is closed. **Most of them close.** That is what the ladder is for, and it is why a monitoring system that only ever escalates is a liability rather than an asset.

On contemporaneous records, correct a common belief: since the **America Invents Act** of 2011 moved the United States to first-to-file, priority contests turn on your **effective filing date**, not on proving you thought of it first. What timestamped technical evidence still buys you:

- **37 CFR 1.130 declarations** — removing a grace-period disclosure from prior art under §102(b)(1). A timestamped commit, a published benchmark, a conference talk. Everyday and non-exotic.
- **Inventorship** — still turns on conception, which the AIA did not touch, and which AI-assisted invention makes sharper.
- **Prior user rights, 35 U.S.C. §273** — worth knowing precisely because it is a **defense**, available to someone accused of infringement, not an offensive right. It requires U.S. commercial use at least a year before the earlier of the filing date or a qualifying disclosure, proved by **clear and convincing evidence**, and it is personal and largely non-transferable. Your version history is what makes it provable. **It does not stop anyone from suing you** — it is what lets you win once you have been, which for most founders is the more realistic exposure. Assert it without a reasonable basis and §273(f) shifts fees against you.

The system is not theoretical; it runs daily against my own portfolio, including in connection with a currently unresolved commercial matter I will not discuss here. On the ladder above that is *executed* — not field validated. It has not run long enough, or in enough hands, to claim the top rung.

Build a Mini-IV

None of the above requires my machinery. It requires the discipline underneath it, and that scales down further than people expect.

A word first about the model I am borrowing from, because I am borrowing from a controversial one.

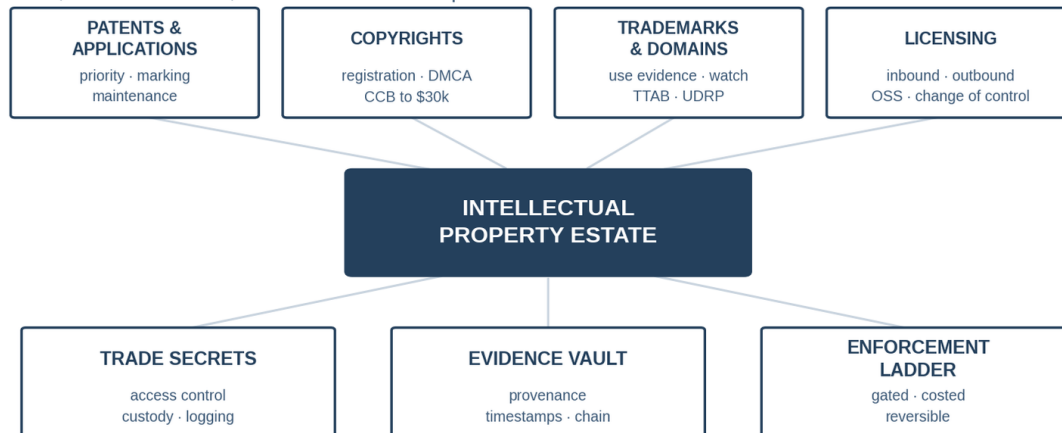
Intellectual Ventures is the largest patent-acquisition business ever built. Founded in 2000 by a former Microsoft chief technology officer and three co-founders — one of them Peter Detkin, the Intel lawyer at the center of the story this article opened with — it raised money from institutional investors, bought patents in bulk, and licensed them to operating companies. Its own Invention Investment Fund states it **invested over \$3 billion in patents acquired across more than a thousand transactions** — that is what the fund spent buying, not what inventors received.

To a great many people in technology it is the archetype of everything wrong with patent assertion, and it has been called the ultimate troll more times than anyone has counted. **I am not endorsing its business model, and you do not have to either.** What is worth taking is narrower and unarguable: at very large expense, it built a systematic function for *knowing what it owned* — which rights it held, what those rights actually covered, which products sat near them, what each was worth, and what to do next about any of it.

That function used to require an institution. **It is the function, not the institution, that I am claiming a founder can now run.** Call it a Mini-IV, after Intellectual Ventures. You do not need the scale. You can borrow the discipline — across the **whole** estate: patents and applications, copyrights, trademarks and domains, **licensing in both directions**, trade secrets, the evidence vault and the enforcement ladder. Not just the patents.

The Mini-IV: operate the whole estate

Not a patent folder. An operating system for every right you hold — each with its own forum, its own remedies, and its own least-cost path.



Do not ask what IP litigation costs. Ask which right is involved, and what its least-cost effective path is.

A trade secret survives only if the information is actually treated as secret. A copyright claim under \$30,000 has a forum that is not federal court. A trademark dispute is not a patent case. And licensing sits across all of them: the terms you grant and the terms you accepted decide what the other six are still worth.

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Figure 7 The Mini-IV — operating the whole estate

One operational warning first, because almost nobody tells founders this. A pro se party has no attorney-client privilege, and work-product protection requires material prepared *in anticipation of litigation*. A registry maintained "as part of development" is an **ordinary-course business record** — **fully discoverable**. If you keep a standing written file of the best arguments against your own validity, and it is not routed through counsel under engagement, you have built your opponent's invalidity exhibit and indexed it.

So: adversarial review, design-around analysis and validity red-teaming go through counsel, under engagement. Everything else can live in the ordinary-course registry. That single structural decision costs almost nothing and is worth more than any tool in this article.

Different rights, different ladders. Copyright exists the moment the work is written down, but you generally cannot file a federal suit for a U.S. work until the Copyright Office has registered it or refused it — and registering *before* the infringement is what unlocks **statutory damages**, a fixed sum you can recover without proving what you lost, plus attorney's fees. (One safe harbor: for a published work, registering within three months of first publication preserves both even against infringement that began earlier — 17 U.S.C. §412.) Copyright also has cheaper routes patent law

lacks: **DMCA notice-and-takedown** (a letter under the Digital Millennium Copyright Act that obliges a host to remove infringing material) and the **Copyright Claims Board**, a small-claims tribunal inside the Copyright Office capped at \$30,000 total, with a \$15,000-per-work cap for timely registered works. The Trademark Trial and Appeal Board — the USPTO body that hears trademark registration disputes — decides registration, not infringement, and awards no damages or injunctions — but under *B&B Hardware v. Hargis Industries*, 575 U.S. 138 (2015), its likelihood-of-confusion findings can bind you in later infringement litigation where the usages adjudicated are materially the same. Trade secrets exist only if you took reasonable measures to keep them secret; 18 U.S.C. §1839 defines one and §1836(b) is the provision you sue under. And **marking** under §287 — physically labelling your product with its patent number, which is what starts the damages clock running against an infringer who did not otherwise know. It does not apply where **only method claims are asserted** (claims to a process rather than a thing, since there is nothing to label), and once an accused infringer meets a low burden of production, **the patent owner bears the burden of proving they complied** — *Arctic Cat v. Bombardier*, 876 F.3d 1350 (Fed. Cir. 2017).

Do not ask what IP litigation costs. Ask **which right is involved, and what its least-cost effective path is.**

The other half of the estate

Everything above treats intellectual property as something you hold and might one day assert. That is half the estate at most.

The other half is what you license, and on what terms — and for a software company it is very often the half that determines whether the rest of it matters.

Licensing is not the exit door from an IP portfolio. It is a control surface *inside* it. Every one of these is a licensing decision with portfolio consequences, and most founders make several of them by accident:

- Which components you release, and under which license — permissive, reciprocal, or not at all
- What your inbound dependencies oblige you to do, and whether anything in your stack has already made that choice for you
- Whether a contributor agreement exists **before** the first outside pull request rather than after
- What a customer contract grants, for how long, and whether it survives your acquisition

- Where a trade secret ends and a public disclosure begins — publishing source code is a disclosure decision with patent consequences, and the **grace period**, the window after your own disclosure in which you can still file, is twelve months in the United States and does not exist at all in much of the rest of the world
- How a commercial license, an open-source release and a patent claim over the same mechanism coexist without one silently defeating the others

My own position, stated the way my published papers state it and no further: **reciprocal open-source terms — the GNU Affero General Public License v3 specifically — remain under consideration for Governed Multi-Route Selection and Constrained Policy Reinforcement Learning, two components of my own system.** *Reciprocal* means anyone who builds on the code must release their changes on the same terms; a *permissive* license would let them take it closed. The reasoning is that a governance layer nobody can inspect is just another promise to trust me, and AGPL rather than a permissive license because that license closes the loophole where someone runs your code as a hosted service and never distributes anything. **No licensing model is committed. Scope, timing and terms are not decided, and plans change.**

The full argument does not fit here, and deserves better than a subsection. **The licensing question — how open-source terms, commercial licenses, contributor agreements, trade secrets and patent rights interact inside one portfolio — is the subject of the next article.** It is the piece of this that founders get wrong most often and most irreversibly, and unlike an enforcement decision, a license you have already granted is very difficult to take back.

The founder's checklist

Enough argument. Ninety days. Most of it is not legal work. None of it requires a litigation budget.

Week 1 — Know what you own

- ☐ **Enumerate every right.** Patents, applications, copyrights, marks, domains, trade secrets, know-how, contracts, licenses. One list. If it does not exist as a list, it does not exist.
- ☐ **Confirm the applicant of record and chain of title on every item.** Founder-held or entity-held? Assignments executed and recorded? This defect is cheap now and fatal later.
- ☐ **Check your employment and contractor agreements** for present-tense assignment language ("hereby assigns," not "agrees to assign"). Anything built before the agreement was signed may not have transferred.

- ☐ **Determine your entity status** — undiscounted, small, or micro. Most seed-stage founders qualify for one of the discounts and are paying full freight.

Weeks 2–4 — Fix the cheap structural things

- ☐ **Register copyrights** in your core codebase and material written works. It gates federal suit and controls statutory damages.
- ☐ **Inventory trade secrets and actually control access.** A list, access controls, NDAs, offboarding, logging. Calling it confidential does not make it a trade secret.
- ☐ **Capture trademark use evidence** — dated specimens, first-use dates, market and goods/services. Evidence of use decays and cannot be reconstructed.
- ☐ **Verify domain ownership** sits with the entity, not a founder's personal account, and is not expiring.

Month 2 — Make the technical record real

- ☐ **Link inventions to code.** Commits, tests, benchmarks, deployments, timestamps. This is a version-control habit, not a legal project.
- ☐ **Write down what a decision required *before* you make it** — the obligation roster habit. It is the single highest-leverage change and it costs nothing.
- ☐ **Run a prior-art sweep** against your own core claims and **write down what you find, including what hurts.**
- ☐ **Route adversarial review through counsel** under engagement, before you generate a single "best argument against us" document.

Month 3 — Operate it

- ☐ **File your most defensible invention with Track One** if timing matters. At \$933 micro / \$1,866 small, a first action lands ~17 months sooner.
- ☐ **Set mechanical deadline control** — priority, maintenance, marking, renewal. Automated, not remembered. Somewhere between half and sixty percent of patentees abandon before full term.
- ☐ **Stand up a competitor watch** on the two or three products nearest your claims.
- ☐ **Write your enforcement ladder before you need it** — the gates, what each costs, what evidence advances you, and what makes you stop.
- ☐ **Decide what you will *not* patent** and keep as a trade secret. That decision is strategy; it is also one of the two on this list that is hard to reverse.

- ☐ **Audit your inbound licenses.** Run a dependency scan and find out what your stack already obliges you to do. Reciprocal terms deep in a dependency tree have decided a company's licensing posture before, without anyone choosing it.
- ☐ **Put a contributor agreement in the repository before it is public,** not after the first pull request.
- ☐ **Check what your customer contracts actually grant** — scope, duration, and whether the grant survives a change of control.

One question to bring to your next board or investor meeting: *what would it cost us to find out whether a competitor crosses our claims?* If nobody knows, that number — not the \$3 million — is what is actually deciding your IP strategy.

Key takeaways

11. **The "\$3 million patent case" is the median total cost, per patent, of a dispute with more than \$25 million at risk, through appeal.** It was never the entry price of caring about a patent.
12. **Almost nothing reaches a courtroom.** About 2.3 percent of terminated cases reach trial; roughly 90 percent end in settlement or on procedural grounds. The visible docket also omits everything resolved before a complaint.
13. **The outcome gap is real; its cause is not established.** Individual inventors win **14.7 percent** of merits decisions against operating companies' 30.6 percent, and 37.0 against 60.9 at trial. That a large share of that gap is procedural rather than substantive — and therefore closeable by operational capacity — is this article's central hypothesis, not its finding.
14. **The vocabulary was manufactured.** "Patent troll" was commissioned by a large company being sued. Works using the term are negative 84 percent of the time; works discussing the same behavior without it, 6 percent.
15. **A micro entity pays twenty cents on the dollar,** and Track One buys a first action ~17 months sooner for \$933. The advice most founders receive predates both.
16. **The asset is cheap because nobody can price it.** Only 24 percent of software startups hold any patent, founders rank it last among protections, 60 percent of holders abandon before term, and quality has been hard to tell apart. Low ownership, no bidders, high abandonment, a noisy signal — that is a description of mispricing, not of a bad asset.
17. **AI raises dispersion, not value** — which is what makes quality legible, and legibility is what reprices an asset class.
18. **Mechanize the checkable, buy the judgment.** Dependent-claim structure and traceability are formal properties. Claim construction and assertion strategy are not.

19. **Closure happens at disclosure time or not at all** — and disclosing without claiming *dedicates to the public*. Exhaustiveness is not a strategy.
 20. **Route adversarial work through counsel** before you create it. An ordinary-course registry of your own doubts is discoverable.
 21. **Licensing is a control surface inside the portfolio, not the exit from it** — and the half where mistakes are hardest to undo. It is the subject of the next piece.
 22. **IP is an instrument, not just an asset**. Design-around distance, procurement diligence, negotiating position and acquisition value are all realised without filing anything. The competitive value of a right lives in the shadow it casts — proportional to how credibly you *could* act, not how often you do.
 23. **The repricing has not shown up in the data yet** — because the data describes cases filed before the tooling existed. That is the window, not a weakness in the case.
 24. **The right existed all along. The friction is what made it safe to ignore** — and the friction is what is changing.
-

Ask yourself

- If I had to prove today what we built and when, what would I actually be able to produce?
 - Which of my rights would survive an adversarial read, and which am I hoping nobody looks at?
 - Am I paying undiscounted fees on a portfolio that qualifies for a 60 or 80 percent discount?
 - Who told me patents do not matter for my company — and what year was their model calibrated?
 - If my strongest patent issued eighteen months sooner, what conversation would it change?
 - What am I deliberately keeping as a trade secret, and can I prove I treated it as one?
 - If someone infringed tomorrow, what is my next lawful step — and do I know it now, or would I be improvising?
 - How many engineering quarters would it cost my nearest competitor to go around what I hold — and has anyone here ever worked that out?
 - Does my sales team know what our IP position lets them say to a procurement officer?
 - What has my dependency tree already committed me to, and did anyone here choose it?
 - If I open-sourced my core component tomorrow, which of my other rights would that strengthen, and which would it quietly extinguish?
-

My position

I am a founder. I have built intellectual property that matters to me, and I do not intend to treat it as passive paperwork.

Every inventor builds on prior work. Learning from prior work is not infringement. Legitimate competition is not infringement. Designing around a patent — building something that does the job without practicing the claim — is not infringement, and the Federal Circuit, the specialist appeals court that hears every patent appeal in the country, has called it "one of the ways in which the patent system works to the advantage of the public." *Slimfold Mfg. v. Kinkead Industries*, 932 F.2d 1453, 1457 (Fed. Cir. 1991).

Independent invention is a different matter, and I will not soften it: **for patents, arriving at the same thing on your own is not a defense.** Copyright works the other way — independent creation is a complete answer there — and the two get conflated constantly. If you build what someone else has claimed, having never seen their patent, you still infringe. That is uncomfortable, it is the law, and it cuts toward me on some days and against me on others.

Respect for the work of others does not require me to surrender the rights attached to my own.

Where the evidence and the economics both justify action, I will act. Where they do not, I will stop — and stop is the outcome most of these gates should produce. That is the discipline, and it applies to me before it applies to anyone else.

None of my tools makes me right. They cannot make an invalid patent valid, convert legitimate competition into infringement, turn public information into a trade secret, make bad evidence admissible, or make a court unnecessary. Rule 11 and 37 CFR 11.18 apply to unrepresented parties too, and require reasonable inquiry before anything is presented. What the tools change is my ability to determine what is true, preserve what matters, name what is still uncertain, and act deliberately.

That is the whole claim, and it is enough. **I am not telling you the fight got easy. I am telling you the price of finding out whether you have one collapsed, and nobody has updated the advice.**

Do not mistake the size of my company for the size of my resolve, or my ability to protect what I have built.

And build your own Mini-IV. Do not file a patent and forget it. Do not call information a trade secret while allowing uncontrolled access. Do not build a brand without preserving evidence of use. Do not wait until infringement occurs to reconstruct years of technical history from memory. Do not trust a language model to remember what you own.

Create the machinery. Maintain the registry. Preserve the evidence. Challenge your own assumptions before someone else does.

Small no longer means undefended.

The large company's bank account has not disappeared. **Its monopoly on institutional capability has.**

Research and legal references

25. Predecessor: *Who Can Afford Friction?* — <https://drive.google.com/file/d/1W4P9zb3yEJSXtCRvLy3hf4edUOCTu70r/view?usp=sharing>
26. AIPLA *Report of the Economic Survey 2015 / 2017 / 2019 / 2023* — patent infringement median costs by amount at risk. The \$3.0M figure: 2017 edition, All Varieties, >\$25M at risk, through trial and appeal.
27. Lex Machina, *Patent Litigation Report 2025* — 10,128 non-ANDA cases terminated 2022–2024.
28. Cotropia, Kesan, Rozema & Schwartz, *Endogenous Litigation Costs*.
29. Joff Wild, IAM, 22 Aug. 2008 (origin of "patent troll," with Peter Detkin's account); Brenda Sandburg, *Trolling for Dollars*, The Recorder, 30 July 2001.
30. Edward Lee, 19 Stan. Tech. L. Rev. 113 (2015) — 570 works; 84% vs 6% negativity.
31. Miller et al., *Who's Suing Us? Decoding Patent Plaintiffs since 2000 with the Stanford NPE Litigation Dataset*, 21 Stan. Tech. L. Rev. 235 (2018) — 10,812 suits, 2000–2015.
32. Allison, Lemley & Schwartz, *How Often Do Non-Practicing Entities Win Patent Suits?*, 32 Berkeley Tech. L.J. 237 (2017) — merits win rates; trial-stage outcomes. Note the small cells behind two of them: universities 30.0% is 3 of 10, failed startups 21.4% is 6 of 28.
33. Unleashing American Innovators Act of 2022, Pub. L. 117-328 Div. W §107; 35 U.S.C. §41(h); 88 FR 17147. USPTO current fee schedule; Track One was established by 76 FR 59050 (23 Sept. 2011); 90 FR 29990 (8 July 2025) raised the annual cap on accepted requests to 20,000.
34. Farre-Mensa, Hegde & Ljungqvist, *What Is a Patent Worth? Evidence from the U.S. Patent "Lottery."*
35. Graham, Merges, Samuelson & Sichelman, *High Technology Entrepreneurs and the Patent System* (Berkeley Patent Survey), 24 Berkeley Tech. L.J. 1255 (2009).
36. Datzov & Rantanen, 110 Iowa L. Rev. 667 (2025) — all 386 Fed. Cir. §101 decisions.
37. Moore, 20 Berkeley Tech. L.J. 1521 (2005) — 53.71% of 1991-issued patents allowed to expire; Crouch, *Patently-O*, 15 Mar. 2026 — the trend since.
38. Lemley, 95 Nw. U. L. Rev. 1495 (2001) — the origin, and the hedge, of the "5% ever litigated or licensed for a royalty" estimate now circulating as "90–95% never licensed."

39. Paul Graham, *Are Software Patents Evil?*, Mar. 2006. Jason Lemkin, SaaStr, 21 Dec. 2017. Steve Blank, First Round Review, 4 Oct. 2024.
40. Magesh et al., *Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools*.
41. MPEP §2161.01; MPEP §2181; *Aristocrat Techs. v. Int'l Game Tech.*, 521 F.3d 1328 (Fed. Cir. 2008); *Williamson v. Citrix Online*, 792 F.3d 1339 (Fed. Cir. 2015) (en banc).
42. 35 U.S.C. §112(d); MPEP 608.01(n); *Pfizer v. Ranbaxy*, 457 F.3d 1284 (Fed. Cir. 2006).
43. 35 U.S.C. §132(a), §120; *Liebel-Flarsheim v. Medrad*, 481 F.3d 1371 (Fed. Cir. 2007); *Johnson & Johnston Assocs. v. R.E. Serv. Co.*, 285 F.3d 1046 (Fed. Cir. 2002) (en banc), as refined by *PSC Computer Products v. Foxconn*, 355 F.3d 1353 (Fed. Cir. 2004).
44. *Aylus Networks v. Apple*, 856 F.3d 1353 (Fed. Cir. 2017); *In re Collect*, 81 F.4th 1216 (Fed. Cir. 2023); *Allergan v. MSN*, 111 F.4th 1358 (Fed. Cir. 2024).
45. USPTO, *Guidance on Use of AI-Based Tools in Practice Before the USPTO*, 89 FR 25609 (11 Apr. 2024); 37 CFR 11.18; 37 CFR 11.5(b)(1); 37 CFR 1.31(a)(1) (91 FR 13519, eff. 20 July 2026); 28 U.S.C. §1654.
46. *Thaler v. Vidal*, 43 F.4th 1207 (Fed. Cir. 2022); USPTO, *Revised Inventorship Guidance for AI-Assisted Inventions*, 90 FR 54636 (28 Nov. 2025).
47. 35 U.S.C. §273; 37 CFR 1.130; 35 U.S.C. §§286, 287; *Arctic Cat v. Bombardier*, 876 F.3d 1350 (Fed. Cir. 2017).
48. 17 U.S.C. §§411–412; U.S. Copyright Office, Copyright Claims Board and §512 resources; *B&B Hardware v. Hargis Indus.*, 575 U.S. 138 (2015); 18 U.S.C. §1836(b), 1839.
49. *Slimfold Mfg. v. Kinkead Indus.*, 932 F.2d 1453 (Fed. Cir. 1991).
50. Harvey — company-reported adoption figures, August 2026.
51. *Lexos Media IP LLC v. Overstock.com*, No. 22-2324-JAR (D. Kan. 2 Feb. 2026) (Robinson, J.) — sanctions for AI-fabricated citations.
52. Lemley & Shapiro, *Probabilistic Patents*, 19 J. Econ. Persp. 75 (2005) — the 1.5% / 0.1% litigation figures.
53. Lee, *Patent Trolls: Moral Panics, Motions in Limine, and Patent Reform*, 19 Stan. Tech. L. Rev. 113 (2015) — 570 **media** works; genre distribution at Part III.
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The right always existed. What protected the conduct around it was the cost of finding out — and that cost is collapsing. The advice you are still getting is twenty years old, and it has moved from fact into myth

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